

Advanced Micro Devices, Inc.

Recommendation **BUY** ★★★★★

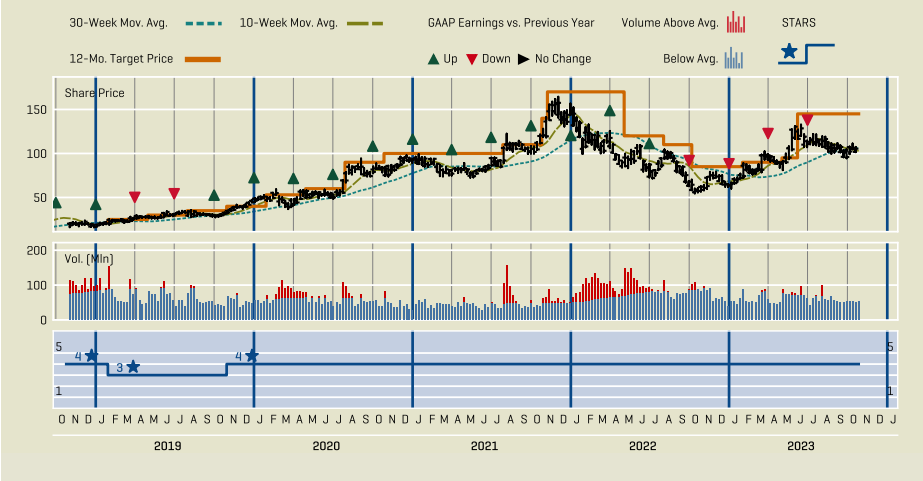
Price USD 96.43 [as of market close Oct 27, 2023] **12-Mo. Target Price** USD 145.00 **Report Currency** USD **Investment Style** Large-Cap Growth

Equity Analyst Angelo Zino, CFA

GICS Sector Information Technology **Sub-Industry** Semiconductors **Summary** Advanced Micro Devices is a leading supplier of microprocessors and graphics semiconductors that are used in computers and related products.

Key Stock Statistics (Source: CFRA, S&P Global Market Intelligence (SPGMI), Company Reports)									
52-Wk Range	USD 132.83 - 58.03	Oper.EPS2023E	USD 2.87	Market Capitalization[B]	USD 155.8	Beta			1.79
Trailing 12-Month EPS	USD 2.54	Oper.EPS2024E	USD 4.41	Yield [%]	N/A	3-yr Proj. EPS CAGR[%]			22
Trailing 12-Month P/E	37.96	P/E on Oper.EPS2023E	33.60	Dividend Rate/Share	N/A	SPGMI's Quality Ranking			B-
USD 10K Invested 5 Yrs Ago	54,697.0	Common Shares Outstg.[M]	1,614.00	Trailing 12-Month Dividend	N/A	Institutional Ownership [%]			72.0

Price Performance



Source: CFRA, S&P Global Market Intelligence
Past performance is not an indication of future performance and should not be relied upon as such.
Analysis prepared by **Angelo Zino, CFA** on Aug 02, 2023 08:48 AM ET, when the stock traded at **USD 117.60**.

Highlights

- ▶ We see sales rebounding 21% in 2024 after our view of a 2.2% decline in 2023 (+44% in 2022). AMD saw lower Q2 CPU demand from both PCs (-54%) and data centers (-11%), while gaming fell by a more modest 4% and Embedded rose 16%. That said, we do believe customer inventories have been sharply curtailed and remain excited about content gains from AI servers. We see greater momentum for AMD's fourth gen EPYC CPUs, nearly doubling sequentially in Q2, as we think AMD's pipeline remains impressive (4 nm Bergamo in 2H; 3 nm Turin in 2024) and should support higher revenue (data center sales +50% in 2H).
- ▶ We estimate margins to expand to 54% in 2024, on more stable PC volume and, compared to our 51% gross margin forecast for 2023 [vs. 52% in 2022]. Over time, we see higher volumes on new product launches (EPYC, Radeon, and Ryzen processors) supporting a better mix. Greater representation from AMD's higher margin Data Center and Embedded segments will also aid margins over time.
- ▶ We view AMD's balance sheet as healthy (\$3.9B net cash) and see sharply improving free cash flow ahead (+\$3.5B forecast in 20223 and rising to +\$6B in 2025). AMD boosted internal inventory in Q2 to support 2H data center ramp.

Investment Rationale/Risk

- ▶ Our Buy rating reflects our growth outlook for AMD's CPU data center servers from the ramp of its next-generation EPYC processor, prospects for GPU offerings (Instinct MI250/MI300 GPUs to ramp in Q4 and '24), and our expectation for balance sheet improvement. Margins will expand in the 2H and for 2024, off cyclical trough, on better mix from new product launches as well as PC-related stabilization/recovery. We positively view AI customer engagements, which grew by more than 7x sequentially in Q2, as we see AMD as a suitable #2 supplier in the data center GPU space (NVIDIA #1). We believe AMD's AI opportunities stem much further than the cloud market given its wide variety of offerings (e.g., CPUs, GPU, FPGAs, and ASICs).
- ▶ Risks to our recommendation and target price include lower demand for computers and servers, greater-than-anticipated pricing pressure, significant market share loss, and slower-than-expected global economic growth.
- ▶ Our 12-month target price of \$145 is based on a P/E of 32.8x our 2024 EPS estimate, above peers. Improving AI prospects, margin expansion potential, attractive net cash/FCF, and greater earnings leverage (\$5-\$6 per share in 2025) should drive multiple expansion.

Analyst's Risk Assessment

LOW	MEDIUM	HIGH
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AMD is subject to the cyclical swings of the semiconductor industry, demand fluctuations for computer end-products, vacillation in average selling prices for chips, and strong competition from Intel, which is a much larger rival in microprocessors.

Revenue/Earnings Data

Revenue (Million USD)					
	1Q	2Q	3Q	4Q	Year
2024	E 6,373	E 6,751	E 7,216	E 7,503	E 27,843
2023	5,353	5,359	E 5,851	E 6,489	E 23,093
2022	3,363	3,629	4,163	4,282	15,438
2021	3,162	3,257	3,481	3,529	13,429
2020	1,786	1,932	2,801	3,244	9,763
2019	1,272	1,531	1,801	2,127	6,731

Earnings Per Share (USD)

	1Q	2Q	3Q	4Q	Year
2024	E 0.91	E 1.05	E 1.20	E 1.25	E 4.41
2023	0.60	0.58	E 0.75	E 0.94	E 2.87
2022	1.13	1.05	0.67	0.69	3.50
2021	0.52	0.63	0.73	0.92	2.79
2020	0.18	0.18	0.41	0.52	1.29
2019	0.06	0.08	0.18	0.32	0.64

Fiscal Year ended Dec 31. EPS Estimates based on CFRA's Operating Earnings; historical earnings are adjusted. In periods where a different currency has been reported, this has been adjusted to match the current quoted currency.

Dividend Data

No cash dividends have been paid in the last year.

Advanced Micro Devices, Inc.

Business Summary Aug 02, 2023

CORPORATE OVERVIEW. Advanced Micro Devices (AMD) designs and sells digital integrated circuits (ICs), including x86 microprocessors and chipsets for computers, embedded microprocessors for commercial and consumer applications, and as a result of the company's acquisition of ATI Technologies in October 2006, graphics processors. Beginning with the second quarter of 2022, AMD changed its segment reporting to the following four segments: Data Center, Client, Gaming, and Embedded.

The Data Center segment primarily includes server microprocessors, GPUs, data processing units (DPUs), Field Programmable Gate Arrays (FPGAs), and adaptive System-on-Chip (SoC) products for data centers. The Client segment primarily includes microprocessors, accelerated processing units (APUs) that integrate microprocessors and graphics, and chipsets for desktop and notebook personal computers. The Gaming segment primarily includes discrete graphics processing units (GPUs), semi-custom SoC products, and development services. The Embedded segment primarily includes embedded microprocessors, GPUs, FPGAs, adaptive SoC products, and Adaptive Compute Acceleration Platform (ACAP) products.

A microprocessor is an IC that serves as the central processing unit (CPU), or brain, of a computer. The performance of a processor is a critical factor for the performance of the computer. The main measures for microprocessor performance include: work-per-cycle, or how many instructions per cycle, clock speed, how fast the CPU's internal logic operates as measured by units of hertz, and power consumption. Other factors impacting performance include the number of cores on a microprocessor, bit rating of the microprocessor, memory size, and data access speed.

A GPU is a programmable logic chip that helps render images, animations, and video and is increasingly being used to handle general computing tasks. GPUs are located in plug-in cards, as a discrete processor or in a chip on the motherboard, or in the same chip as the CPU as part of an APU or SoC. GPUs on stand-alone cards or discrete GPUs on the motherboard typically access their own memory, while GPUs in the chipset or CPU chip share main memory with the CPU.

A server is a computer system that performs services for connected customers as part of a client-server architecture. Many servers are designed to run an application or applications, often for extended periods of time with minimal human intervention. Examples of servers include cloud, web, e-mail, print, and on-premise servers. Servers are also used in cloud computing, which is a computing model where data, applications, and services are delivered over the internet or an intranet which can be rapidly provisioned and released with minimal effort.

Embedded products address computing needs in enterprise-class telecommunications, networking, security, and storage systems. Typically, AMD embedded products are used in applications that require high to moderate levels of performance, where key features may include relatively low power, small form factor, and 24x7 operations.

AMD also leverages its core IP, including its graphics and processing technologies, to develop semi-custom solutions for the PC and gaming markets. In this market, semiconductor suppliers work alongside system designers and manufacturers to enhance the performance and overall user experience for semi-custom customers. AMD has used this type of collaborative co-development approach with many of today's leading game console manufacturers and can also address customer needs in many other markets beyond game consoles.

CORPORATE STRATEGY. The microprocessor market is a critical part of the semiconductor industry, and is dominated by two companies, Intel and AMD. The two competitors have battled for preeminence in the segment for decades. Intel now ships over 80% of the world's microprocessors, and is still the clear leader in this space. It has accomplished this by extending its leadership in key technologies that have provided important competitive advantages. Meanwhile, AMD accounts for less than 20% of the microprocessor market and has made many changes over the years to build a sustainable business model.

IMPACT OF MAJOR DEVELOPMENTS. In February 2022, AMD acquired XLNX in an all-stock transaction valued at over \$50 billion. XLNX shareholders received 1.7234 shares of AMD stock per XLNX share, which equated to about 26% ownership in the combined company for XLNX investors. We view the transaction positively from the perspective that brings together complementary product offerings (CPUs and GPUs from AMD; FPGAs from XLNX). We note that XLNX also increases AMD's addressable market, with opportunities from the emergence of edge computing (e.g., automotive). The deal was immediately accretive.

FINANCIAL TRENDS. AMD's revenues are highly cyclical and tend to move with the broader semiconductor market. We note that there are industry-specific factors, such as average selling prices and sales mix, that have also caused variability in annual sales results. The swings in sales have caused wide changes in margins because of the company's fixed cost structure that included plants and equipment, and when sales dropped with the broader economy during the last recession, losses ensued. However, we see a steadier margin profile following the spin-off of its manufacturing arm in 2009.

Revenue of \$23.6 billion grew 44% in 2022 driven by higher Embedded, Data Center, and Gaming segment revenue, partially offset by lower Client segment revenue. Pro forma revenue rose 20% on a combined AMD and Xilinx company basis. Non-GAAP gross margin widened to 52% from 48% in 2022, on a richer product mix with higher Embedded and Data Center revenue, partially offset by lower Client revenue.

As of the end of June 2023, AMD's debt obligations were \$2.5 billion, while cash was \$6.4 billion. We anticipate sharply higher free cash flow generation ahead, estimating more than \$5 billion in 2024, which compares to the \$3.1 billion posted in 2022. Despite near-term uncertainty, we see improving results across AMD's business, specifically related to planned new product releases, supporting a stronger balance sheet in the coming years.

Corporate information

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D. M. Smith	H. A. Wolin

CTO and Executive VP of Technology & Engineering	Senior Vice President of Global Operations
M. D. Papermaster	K. Keshvari

Executive VP, CFO & Treasurer	Chair & CEO
J. X. Hu	L. T. Su

President
V. Peng

Board Members	
A. Y. Talwalkar	L. T. Su
E. W. Vanderslice	M. D. Durcan
J. A. Householder	M. P. Gregoire
J. Olson	N. M. Denzel
J. W. Marren	

Domicile	Auditor
Delaware	Ernst & Young LLP

Founded
1969

Employees
25,000

Stockholders
5,014



Advanced Micro Devices, Inc.

Quantitative Evaluations							Expanded Ratio Analysis				
Fair Value Rank	NR	1	2	3	4	5		2022	2021	2020	2019
		LOWEST					HIGHEST				
		Based on CFRA's proprietary quantitative model, stocks are ranked from most overvalued [1] to most undervalued [5].									
Fair Value Calculation	N/A						Price/Sales	4.31	10.93	11.35	7.68
							Price/EBITDA	21.65	44.29	65.92	60.63
							Price/Pretax Income	84.94	48.87	86.57	139.04
							P/E Ratio	18.51	52.38	71.17	72.16
							Avg. Diluted Shares Outstg. [M]	1,571.00	1,229.00	1,207.00	1,120.00
							Figures based on fiscal year-end price				
Volatility		LOW		AVERAGE		HIGH					
Technical Evaluation	NEUTRAL	Since July, 2023, the technical indicators for AMD have been NEUTRAL "					Key Growth Rates and Averages				
Insider Activity		UNFAVORABLE		NEUTRAL		FAVORABLE	Past Growth Rate [%]		1 Year	3 Years	5 Years
							Net Income		NM	57.01	109.13
							Sales		43.61	51.92	35.05
							Ratio Analysis (Annual Avg.)				
							Net Margin [%]		5.59	16.78	12.12
							% LT Debt to Capitalization		4.28	3.16	13.52
							Return on Equity [%]		4.24	36.38	32.40

Company Financials Fiscal year ending Dec 31											
Per Share Data (USD)		2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
Tangible Book Value		3.78	5.70	4.39	1.99	0.75	0.07	-0.11	-1.11	-0.54	-0.51
Free Cash Flow		2.00	2.65	0.66	0.25	-0.13	-0.11	0.00	-0.41	-0.25	-0.31
Earnings		0.84	2.57	2.06	0.30	0.32	-0.03	-0.60	-0.84	-0.53	-0.11
Earnings (Normalized)		3.50	2.79	1.29	0.64	0.46	0.17	-0.14	-0.54	0.06	-0.11
Dividends		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Payout Ratio (%)		NM	NM	NM	NM	NM	NM	NM	NM	NM	NM
Prices: High		152.42	164.46	97.98	47.31	34.14	15.65	12.42	3.37	4.80	4.65
Prices: Low		54.57	72.50	36.75	16.94	9.04	9.42	1.75	1.61	2.35	2.25
P/E Ratio: High		43.50	58.90	76.00	73.90	74.20	92.10	NM	NM	80.00	NM
P/E Ratio: Low		15.60	26.00	28.50	26.50	19.70	55.40	NM	NM	39.20	NM
Income Statement Analysis (Million USD)											
Revenue		23,601	16,434	9,763	6,731	6,475	5,253	4,319	3,991	5,506	5,299
Operating Income		1,264	3,648	1,369	631.00	451.00	127.00	-383.00	-352.00	163.00	85.00
Depreciation + Amortization		3,524	463.00	354.00	258.00	170.00	144.00	133.00	167.00	190.00	223.00
Interest Expense		88.00	34.00	47.00	94.00	121.00	126.00	156.00	160.00	177.00	177.00
Pretax Income		1,198	3,675	1,280	372.00	328.00	-15.00	-459.00	-646.00	-398.00	-74.00
Effective Tax Rate		-10.20	14.00	-94.50	8.30	-2.70	-120.00	-8.50	-2.20	-1.30	-12.20
Net Income		1,320	3,162	2,490	341.00	337.00	NM	NM	NM	NM	NM
Net Income (Normalized)		787.50	2,266	832.50	341.90	212.50	NM	NM	NM	NM	NM
Balance Sheet and Other Financial Data (Million USD)											
Cash		5,855	3,608	2,290	1,503	1,156	1,185	1,264	785.00	1,040	1,097
Current Assets		15,019	8,583	6,143	4,597	3,540	2,634	2,530	2,320	2,736	2,884
Total Assets		67,580	12,419	8,962	6,028	4,556	3,552	3,321	3,084	3,767	4,337
Current Liabilities		6,369	4,240	2,417	2,359	1,984	1,513	1,346	1,403	1,440	1,618
Long Term Debt		2,467	1.00	330.00	486.00	1,114	1,325	1,435	2,007	2,028	1,987
Total Capital		57,706	8,158	6,368	3,512	2,516	1,991	1,851	1,825	2,399	2,602
Capital Expenditures		450.00	301.00	294.00	217.00	163.00	113.00	77.00	96.00	95.00	84.00
Cash from Operations		3,565	3,521	1,071	493.00	34.00	12.00	81.00	-226.00	-98.00	-148.00
Current Ratio		2.36	2.02	2.54	1.95	1.78	1.74	1.88	1.65	1.90	1.78
% Long Term Debt of Capitalization		4.30	0.00	5.20	13.80	44.30	66.50	77.50	110.00	84.50	76.40
% Net Income of Revenue		5.60	19.20	25.50	5.10	5.20	-0.60	-11.50	-16.50	-7.30	-1.60
% Return on Assets		1.98	21.33	11.42	7.45	6.95	2.31	-7.47	-6.42	2.51	1.27
% Return on Equity		4.20	47.40	57.50	16.70	36.20	-6.50	NM	586.70	NM	-15.30

Source: S&P Global Market Intelligence. Data may be preliminary or restated; before results of discontinued operations/special items. Per share data adjusted for stock dividends; EPS diluted. E-Estimated. NA-Not Available. NM-Not Meaningful. NR-Not Ranked. UR-Under Review.

Advanced Micro Devices, Inc.

Sub-Industry Outlook

We have a positive fundamental outlook for the semiconductors sub-industry over the next 12 months. We expect semiconductor sales to rebound about 10% in 2024 after our forecast for a 12% decline in 2023, with a cyclical recovery to start in the second half of the year [Y/Y revenue growth returns in Q4 2023]. We are starting to see a return to sequential growth across the more cyclical areas of the semiconductor market (PCs, smartphones, and gaming), driven in part by seasonality as well as a more normalized inventory channel after sharply undershipping end-demand for the last several quarters. Whether or not recent strength can be sustained will be dependent on macro factors and the health of the consumer, but we think the industry is in a very healthy situation.

Currently, the biggest question mark is within the industrial/automotive markets as those areas held up well through 1H 2023, but channel inventory is now being rightsized due to end-market softness, which may continue until at least Q1 2024. Although inventories are still above optimal levels, companies are making efforts (e.g., production cuts) to quickly get back to normalized levels. In the long term, we expect content growth in automobiles to sharply outpace semiconductor industry growth, driven by several megatrends, such as autonomy, electrification, and connected infotainment. Analog semiconductors are particularly exposed to these trends.

Although data center spending has been relatively muted in 2023 (we estimate ~2%), we see a 10%-15% rebound in 2024, with investments continuing to be focused on upgrading the network to meet more artificial intelligence (AI) driven workloads. As a result, we expect accelerated computing technologies (GPUs, FPGAs, and ASICs) to gain traction, in our view, as they help overcome the limitations of CPUs. We believe that the recent spend shift towards accelerators will be permanent and will grow to an annual run rate of over \$130 billion by 2027 versus our estimated \$45 billion-\$50 billion run rate in 2023. Chipmakers tied to the data center are exposed to the best financially positioned

customers on the planet, and despite quarterly lumpiness, the long-term prospects related to themes like AI and the metaverse remain extremely encouraging and offer significant growth potential ahead.

We believe the semiconductor industry will witness a super cycle over the next decade as content growth per device across a host of markets accelerates, driven by the emergence of greater AI capabilities. Given this, we estimate the semiconductors industry will grow at a 10-year annualized pace of 6%-7% (using 2021 expected revenue as the base year), or at least 2x global GDP.

While companies that outsource manufacturing would typically have more stability in gross margins (a trade-off for capped upside), others with their own manufacturing see more variability. Industry margins are a function of manufacturing utilization and inventory supply-demand imbalances. We forecast some companies, especially in analog, to maintain higher inventory levels. Healthy free cash flow generation and high cash levels will likely result in greater shareholder return via share repurchases. Regulatory and geopolitical uncertainties remain notable risks.

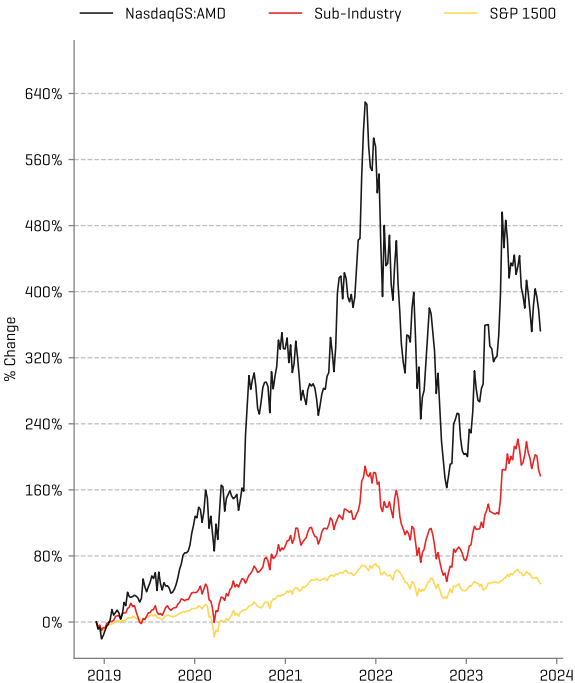
YTD through September 22, the S&P semiconductors sub-industry index rose 64.0% vs. an 11.6% increase for the S&P 1500 index. For full-year 2022, the S&P semiconductors sub-industry index fell 37.9% vs. a 19.1% decline for the S&P 1500 index.

/ Angelo Zino, CFA

Industry Performance

GICS Sector: Information Technology
Sub-Industry: Semiconductors

Based on S&P 1500 Indexes
Five-Year market price performance through Oct 28, 2023



NOTE: A sector chart appears when the sub-industry does not have sufficient historical index data.
All Sector & Sub-Industry information is based on the Global Industry Classification Standard (GICS).
Past performance is not an indication of future performance and should not be relied upon as such.
Source: CFRA, S&P Global Market Intelligence

Sub-Industry: Semiconductors Peer Group*: Semiconductors												
Peer Group	Stock Symbol	Exchange	Currency	Recent Stock Price	Stk. Mkt. Cap. (M)	30-Day Price Chg. (%)	1-Year Price Chg. (%)	P/E Ratio	Fair Value Calc.	Yield (%)	Return on Equity (%)	LTD to Cap (%)
Advanced Micro Devices, Inc.	AMD	NasdaqGS	USD	93.67	151,340.0	-2.4	56.8	37.0	N/A	N/A	-0.0	3.0
Analog Devices, Inc.	ADI	NasdaqGS	USD	160.86	80,159.0	-6.7	13.8	15.0	154.34	2.1	10.3	15.0
Arm Holdings plc	ARM	NasdaqGS	USD	50.17	51,478.0	-6.3	N/A	129.0	N/A	N/A	N/A	N/A
Intel Corporation	INTC	NasdaqGS	USD	32.52	136,194.0	-3.9	19.5	42.0	N/A	1.5	-0.9	30.2
Marvell Technology, Inc.	MRVL	NasdaqGS	USD	46.63	40,232.0	-9.8	20.3	28.0	N/A	0.5	-2.4	15.8
Microchip Technology Incorporated	MCHP	NasdaqGS	USD	72.76	39,606.0	-4.3	16.3	12.0	74.86	2.3	37.4	35.9
Micron Technology, Inc.	MU	NasdaqGS	USD	64.53	70,856.0	-5.0	16.4	NM	N/A	0.7	-12.4	20.5
NXP Semiconductors N.V.	NXPI	NasdaqGS	USD	179.56	46,291.0	-7.9	21.9	13.0	N/A	2.3	35.8	51.8
QUALCOMM Incorporated	QCOM	NasdaqGS	USD	105.62	117,872.0	-3.7	-10.6	11.0	132.90	3.0	47.0	40.2
STMicroelectronics N.V.	STM	NYSE	USD	40.25	37,112.0	-3.7	16.4	9.0	N/A	0.6	35.9	13.4
Texas Instruments Incorporated	TXN	NasdaqGS	USD	144.01	130,790.0	-8.8	-8.8	19.0	138.71	3.6	45.6	39.2

*For Peer Groups with more than 10 companies or stocks, selection of issues is based on market capitalization.
NA-Not Available; NM-Not Meaningful.
Note: Peers are selected based on Global Industry Classification Standards and market capitalization. The peer group list includes companies with similar characteristics, but may not include all the companies within the same industry and/or that engage in the same line of business.

Advanced Micro Devices, Inc.

Analyst Research Notes and other Company News

August 02, 2023

06:55 AM ET... CFRA Maintains Buy Opinion on Shares of Advanced Micro Devices [AMD 117.60****]:

We keep our 12-month target at \$145, on P/E of 32.8x our 2024 EPS view, below historical but above peers. We keep our 2023 EPS estimate at \$2.87 and 2024's at \$4.41. AMD posts Q2 EPS of \$0.58 vs. \$1.05, beating the \$0.57 consensus view. Sales fell 18%, beating expectations, as lower CPU demand hurt both PCs [-54%] and data centers [-11%], while gaming fell by a more modest 4% and Embedded rose 16%. Gross margin compressed 400 basis points to 50% but widened sequentially, as we think PC sales have bottomed and see 2H expansion. We positively view AI customer engagements, up more than 7x sequentially, and will support demand for AMD's Instinct MI250/MI300 GPUs in Q4 and for '24. We see share gain from AMD's fourth gen EPYC CPUs, nearly doubling from Q1, with upside from Bergamo and Genoa X. We do believe customer inventories have been sharply curtailed and remain excited about content gains from AI servers. We see margin expansion off cyclical trough like FCF potential (+\$3.5B in '23 rising to +\$6B in '25). / Angelo Zino, CFA

June 08, 2023

03:13 PM ET... CFRA Maintains Buy Opinion on Shares of Advanced Micro Devices, Inc. [AMD 121.36****]:

We boost our 12-month target price to \$145 from \$95, on a higher revised P/E of 32.8x our 2024 EPS view, below historical but above peers. We maintain our 2023 EPS estimate at \$2.87 but raise 2024's to \$4.41 from \$4.14. We expect revenue to officially trough in Q2, with a sharp inflection to be realized in 2H [data center sales +50% vs. 1H], as excess CPU inventories are fully digested. We see momentum supported by AMD's next-gen data center CPUs, with Bergamo [4nm] in 2H and Turin in 2024 [3nm]. Also, AMD's Instinct MI300 processor [combines GPU/CPU cores] will see significant interest from cloud providers and vie with NVIDIA's Grace Hopper Superchip. AMD's stronger partnership with Microsoft should also drive upside, as it is reportedly cited to be developing a custom processor chip that Microsoft will use for AI workloads. Improving AI prospects, margin expansion off cyclical trough, attractive net cash/FCF, and greater earnings leverage [\$5-\$6 per share in 2025] should drive multiple expansion. / Angelo Zino, CFA

May 04, 2023

03:57 PM ET... CFRA Maintains Buy Opinion on Shares of Advanced Micro Devices, Inc. [AMD 87.57****]:

According to an unconfirmed Bloomberg report, Microsoft [MSFT ****] is working with AMD on the expansion into AI processors. It appears as if Microsoft is providing financial support to AMD to develop a custom processor chip that Microsoft will use for AI workloads, code-named Athena. Despite limited details, we view it as an overall positive for AMD as it bolsters the thesis that the company will be among a host of winners within the semiconductor industry, as we move towards a more AI-enabled world. That said, we would highlight that this deal is unlikely to have an impact on AMD's financial results over the next 12 months, but could further solidify the relationship the two parties have across other offerings. Also, we would not view the news as a negative for peer NVIDIA [NVDA ****]. In fact, we view it as a positive for NVDA as it likely indicates that demand for GPUs and other AI-enabled chips [e.g., ASICs] is very strong, and to some extent, supply-constrained given the needs of cloud providers. / Angelo Zino, CFA

May 03, 2023

08:03 AM ET... CFRA Maintains Buy Opinion on Shares of Advanced Micro Devices [AMD 83.60****]:

We raise our 12-month target to \$95 from \$90, on P/E of 23x our '24 EPS view, below historical but above peers. We reduce our '23 EPS estimate to \$2.87 from \$3.51 and '24 to \$4.14 from \$4.50. AMD posts Q1 EPS of \$0.60 vs. \$1.13, beating the \$0.56 consensus view. Sales fell 9%, near expectations, on declines in Client [-65%] and gaming [-6%], partly offset by the contribution of Xilinx within Embedded. Data Center sales were flat, as higher EPYC sales were offset by enterprise declines. Although we acknowledge EPYC/data center share gains are likely to slow as Intel ramps next-gen servers [3nm in '24], we think AMD's pipeline remains impressive [4nm Bergamo in 2H and 3nm Turin in '24] and should support sharply higher revenue ahead [data center sales +50% in 2H]. We think gross margins [compressed 270 basis points to 50%] and PC orders have bottomed but are cautious about the timing/magnitude of recovery for both. We note net cash of \$3.5B [\$5.9B gross] and see FCF sharply improving from trough levels. / Angelo

Zino, CFA

February 01, 2023

06:25 AM ET... CFRA Maintains Buy Opinion on Shares of Advanced Micro Devices [AMD 75.15****]:

We raise our 12-month target to \$90 from \$85, on P/E of 20x our '24 EPS view, below historical but above peers. We reduce our '23 EPS estimate to \$3.51 from \$3.71 but keep '24 at \$4.50. AMD posts Q4 EPS of \$0.69 vs. \$0.92, beating the \$0.67 consensus view. Sales grew 16%, led by growth within AMD's Embedded [up 19x on Xilinx] and Data Center [+42%] segments, partly offset by lower Client [-51% due to weak PCs] and Gaming [-7%] demand. We are encouraged by cloud adoption for its fourth generation EPYC processors, but see enterprise declines on macro headwinds. We like sharply lower sell-in vs. sell-through [lower channel inventory] as well as new product launches, which should allow PC revenue to trough in Q1. While gross margin of 51% is below the mid '22 peak of 54% [further compression to 50% in Q1], we see expansion in the 2H on better mix and Client recovery. We see CPU server share gain and AI supporting new growth opportunities. We note net cash of \$3.4B and see FCF generation of about \$5B in '23. / Angelo Zino, CFA

November 02, 2022

05:55 AM ET... CFRA Maintains Buy Opinion on Shares of Advanced Micro Devices [AMD 59.66****]:

We keep our 12-month target at \$85, shifting to a P/E of 18.9x our '24 EPS view, which we start at \$4.50, below historical but above peers to reflect share gain. We cut our '22 EPS estimate to \$3.55 from \$4.00 and '23 to \$3.71 from \$4.23. AMD posts Q3 EPS of \$0.67 vs. \$0.73, missing the \$0.70 consensus view. Sales grew 29%, near lower revised preliminary figures, as data center, gaming, and Embedded growth was partly offset by lower PC revenue. Gross margin expanded to 50% from 48% [below original guidance of 54%], on better mix but hurt by inventory charges. We do think lower PC and gaming related revenue is providing additional downside in Q4/Q1, as AMD looks to clear any internal/channel excess inventory. Good visibility in data centers and Embedded is providing some stability for revenue/margins. We think AMD continues to take share in the server CPU space while the launch of AMD's next generation 5nm data center and graphics products will support revenue. Geopolitical tensions with China are a risk. / Angelo Zino, CFA

October 07, 2022

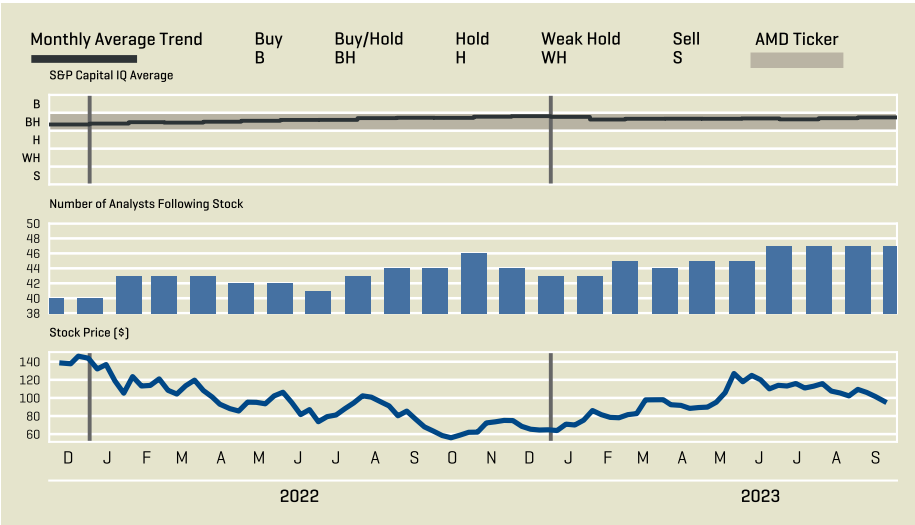
06:28 AM ET... CFRA Maintains Buy Opinion on Shares of Advanced Micro Devices [AMD 67.85****]:

We reduce our 12-month target price to \$85 from \$110 on a P/E of 20x our '23 EPS view, below historical but above peers to reflect share gain. We cut our '22 EPS estimate to \$4.00 from \$4.38 and '23 to \$4.23 from \$4.88. AMD provides preliminary Q3 results and now sees revenue growth of 29% to \$5.6B, well below prior of \$6.7B, on lower client revenue [-40%] due to a sharply declining PC market/inventory correction. Revenue from data center [+45%], gaming [+14%], and embedded [up more than 15x with Xilinx] were all resilient. Gross margin also disappointed [50% vs. 54% expectation], on lower processor units and average selling prices. We also note \$160M in expected CPU/GPU write-downs. Although we are not surprised by the lower guidance, given recent PC trends and commentary from peers, we are disappointed by the magnitude and fear the macro weakness will trickle into other segments [e.g., data centers]. We are optimistic of new product cycles [5nm], cloud share gain, and balance sheet strength/FCF potential. / Angelo Zino, CFA

Note: Research notes reflect CFRA's published opinions and analysis on the stock at the time the note was published. The note reflects the views of the equity analyst as of the date and time indicated in the note, and may not reflect CFRA's current view on the company.

Advanced Micro Devices, Inc.

Analysts Recommendations



	No. of Recommendations	% of Total	1 Mo.Prior	3 Mos.Prior
Buy	25	53	25	24
Buy/Hold	7	15	7	6
Hold	14	30	14	16
Weak hold	0	0	0	0
Sell	0	0	0	0
No Opinion	1	2	1	1
Total	47	100	47	47

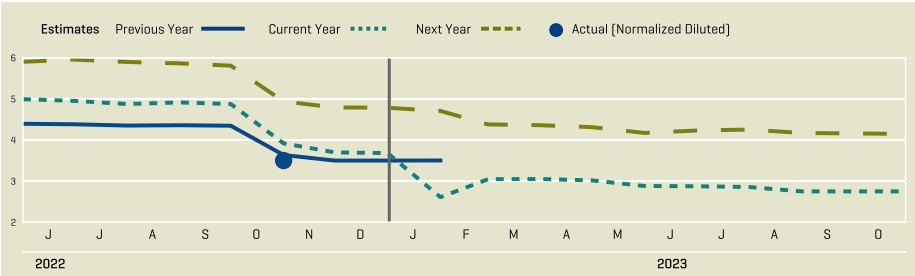
Wall Street Consensus Opinion

Buy/Hold

Wall Street Consensus vs. Performance

For fiscal year 2023, analysts estimate that AMD will earn USD 2.75. For fiscal year 2024, analysts estimate that AMD's earnings per share will grow by 50.85% to USD 4.15.

Wall Street Consensus Estimates



Fiscal Year	Avg Est.	High Est.	Low Est.	# of Est.	Est. P/E
2024	4.15	6.25	3.45	41	22.58
2023	2.75	2.92	2.61	39	34.06
2024 vs. 2023	▲ 51%	▲ 114%	▲ 32%	▲ 5%	▼ -34%
Q3'24	1.11	1.67	0.90	26	84.69
Q3'23	0.68	0.71	0.57	35	138.55
Q3'24 vs. Q3'23	▲ 64%	▲ 134%	▲ 58%	▼ -26%	▼ -39%

Forecasts are not reliable indicator of future performance.
Note: A company's earnings outlook plays a major part in any investment decision. S&P Global Market Intelligence organizes the earnings estimates of over 2,300 Wall Street analysts, and provides their consensus of earnings over the next two years, as well as how those earnings estimates have changed over time. Note that the information provided in relation to consensus estimates is not intended to predict actual results and should not be taken as a reliable indicator of future performance.
Note: For all tables, graphs and charts in this report that do not cite any reference or source, the source is S&P Global Market Intelligence.

Advanced Micro Devices, Inc.

Glossary

STARS

Since January 1, 1987, CFRA Equity and Fund Research Services, and its predecessor S&P Capital IQ Equity Research has ranked a universe of U.S. common stocks, ADRs [American Depositary Receipts], and ADSs [American Depositary Shares] based on a given equity's potential for future performance. Similarly, we have ranked Asian and European equities since June 30, 2002. Under proprietary STARS [Stock Appreciation Ranking System], equity analysts rank equities according to their individual forecast of an equity's future total return potential versus the expected total return of a relevant benchmark [e.g., a regional index (MSCI AC Asia Pacific Index, MSCI AC Europe Index or S&P 500® Index)], based on a 12-month time horizon. STARS was designed to help investors looking to put their investment decisions in perspective. Data used to assist in determining the STARS ranking may be the result of the analyst's own models as well as internal proprietary models resulting from dynamic data inputs.

S&P Global Market Intelligence's Quality Ranking

[also known as **S&P Capital IQ Earnings & Dividend Rankings**] - Growth and S&P Capital IQ Earnings & Dividend Rankings stability of earnings and dividends are deemed key elements in establishing S&P Global Market Intelligence's earnings and dividend rankings for common stocks, which are designed to capsule the nature of this record in a single symbol. It should be noted, however, that the process also takes into consideration certain adjustments and modifications deemed desirable in establishing such rankings. The final score for each stock is measured against a scoring matrix determined by analysis of the scores of a large and representative sample of stocks. The range of scores in the array of this sample has been aligned with the following ladder of rankings:

A+ Highest	B Below Average
A High	B- Lower
A Above	C Lowest
B+ Average	D In Reorganization
NC Not Ranked	

EPS Estimates

CFRA's earnings per share (EPS) estimates reflect analyst projections of future EPS from continuing operations, and generally exclude various items that are viewed as special, non-recurring, or extraordinary. Also, EPS estimates reflect either forecasts of equity analysts; or, the consensus [average] EPS estimate, which are independently compiled by S&P Global Market Intelligence, a data provider to CFRA. Among the items typically excluded from EPS estimates are asset sale gains; impairment, restructuring or merger-related charges; legal and insurance settlements; in process research and development expenses; gains or losses on the extinguishment of debt; the cumulative effect of accounting changes; and earnings related to operations that have been classified by the company as discontinued. The inclusion of some items, such as stock option expense and recurring types of other charges, may vary, and depend on such factors as industry practice, analyst judgment, and the extent to which some types of data is disclosed by companies.

12-Month Target Price

The equity analyst's projection of the market price a given security will command 12 months hence, based on a combination of intrinsic, relative, and private market valuation metrics, including Fair Value.

Abbreviations Used in Equity Research Reports

- CAGR - Compound Annual Growth Rate
- CAPEX - Capital Expenditures
- CY - Calendar Year
- DCF - Discounted Cash Flow
- DDM - Dividend Discount Model
- EBIT - Earnings Before Interest and Taxes
- EBITDA - Earnings Before Interest, Taxes, Depreciation & Amortization
- EPS - Earnings Per Share
- EV - Enterprise Value
- FCF - Free Cash Flow
- FFO - Funds From Operations
- FY - Fiscal Year
- P/E - Price/Earnings
- P/NAV - Price to Net Asset Value
- PEG Ratio - P/E-to-Growth Ratio
- PV - Present Value
- R&D - Research & Development
- ROCE - Return on Capital Employed
- ROE Return on Equity
- ROI - Return on Investment
- ROIC - Return on Invested Capital
- ROA - Return on Assets
- SG&A - Selling, General & Administrative Expenses
- SOTP - Sum-of-The-Parts
- WACC - Weighted Average Cost of Capital

Dividends on American Depositary Receipts (ADRs) and American Depositary Shares (ADSs) are net of taxes (paid in the country of origin).

Qualitative Risk Assessment

Reflects an equity analyst's view of a given company's operational risk, or the risk of a firm's ability to continue as an ongoing concern. The Qualitative Risk Assessment is a relative ranking to the U.S. STARS universe, and should be reflective of risk factors related to a company's operations, as opposed to risk and volatility measures associated with share prices. For an ETF this reflects on a capitalization-weighted basis, the average qualitative risk assessment assigned to holdings of the fund.

STARS Ranking system and definition:

★★★★★ 5-STARS (Strong Buy):

Total return is expected to outperform the total return of a relevant benchmark, by a notable margin over the coming 12 months, with shares rising in price on an absolute basis.

★★★★★ 4-STARS (Buy):

Total return is expected to outperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 3-STARS (Hold):

Total return is expected to closely approximate the total return of a relevant benchmark over the coming 12 months.

★★★★★ 2-STARS (Sell):

Total return is expected to underperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 1-STAR (Strong Sell):

Total return is expected to underperform the total return of a relevant benchmark by a notable margin over the coming 12 months, with shares falling in price on an absolute basis.

Relevant benchmarks:

In North America, the relevant benchmark is the S&P 500 Index, in Europe and in Asia, the relevant benchmarks are the MSCI AC Europe Index and the MSCI AC Asia Pacific Index, respectively.



Advanced Micro Devices, Inc.

Disclosures

Stocks are ranked in accordance with the following ranking methodologies:

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Quantitative Stock Reports:

Quantitative rankings are determined by ranking a universe of common stocks based on 5 measures or model categories: Valuation, Quality, Growth, Street Sentiment, and Price Momentum. In the U.S., a sixth sub-category for Financial Health will also be displayed. Percentile scores are used to compare each company to all other companies in the same universe for each model category. The five [six] model category scores are then weighted and rolled up into a single percentile ranking for that company. For reports containing quantitative rankings refer to the Glossary section seof the report for detailed methodology and the definition of Quantitative rankings.

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STARS Stock Reports:

Global STARS Distribution as of September 30, 2023

Ranking	North America	Europe	Asia	Global
Buy	36.8%	34.5%	41.6%	37.4%
Hold	53.8%	51.2%	50.3%	52.6%
Sell	9.4%	14.3%	8.1%	10.1%
Total	100.0%	100.0%	100.0%	100.0%

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